Association between organic dietary choice during pregnancy and hypospadias in offspring: a study of mothers of 306 boys operated on for hypospadias.

PURPOSE: The etiology of hypospadias is poorly understood. Exposure to pesticides has been considered a risk factor, although findings are inconsistent. Diet constitutes a significant exposure route for pesticides, and pesticide residues are more frequently reported in conventional than organic food products. We examined the association between organic dietary choice during pregnancy and presence of hypospadias in the offspring. MATERIALS AND METHODS: Mothers of 306 boys operated on for hypospadias were frequency matched for geography and child birth year to 306 mothers of healthy boys in a case-control study. Telephone interviews were conducted regarding demographic and lifestyle factors, including intake and organic choice of selected food items (milk, dairy products, egg, fruit, vegetables and meat). Logistic regression models were constructed for dietary variables, and odds ratios were calculated controlling for maternal age, body mass index and alcohol consumption. RESULTS: Overall organic choice of food items during pregnancy was not associated with hypospadias in the offspring. However, frequent current consumption of high fat dairy products (milk, butter) while rarely or never choosing the organic alternative to these products during pregnancy was associated with increased odds of hypospadias (adjusted OR 2.18, 95% CI 1.09-4.36). CONCLUSIONS: This large case-control study of boys operated on for hypospadias suggests an association between hypospadias in the offspring and the mother not choosing the organic alternative, and having a high current intake of nonorganic butter and cheese. This finding could be due to chemical contamination of high fat dairy products. However, general lifestyle and health behavior related to choosing organic alternatives may also explain the finding.